```
View(airquality)
head(airquality)        # First few rows
summary(airquality)     # Summary showing NA counts
is.na(airquality)       # TRUE/FALSE for each cell
anyNA(airquality)       # Checks if any NA exists
sum(is.na(airquality))  # Total number of missing values
colSums(is.na(airquality))   # Count of NAs per column
sapply(airquality, function(x) sum(is.na(x)))  # Alternative
which(is.na(airquality))   #index positions (in vector form)
na.omit(airquality)        # Remove rows with any NA
#Replace NA in Ozone with the column mean (ignoring NAs)
airquality$Ozone[is.na(airquality$Ozone)] <- mean(airquality$Ozone, na.rm = TRUE)
airquality$Ozone    # NA's are replaced
View(airquality)


# Make a copy of the dataset
aq <- airquality

# Convert 'Ozone' to character so we can insert blanks (NA can stay in numeric)
aq$Ozone <- as.character(aq$Ozone)

#create a new data with empty string and space
# Introduce empty strings and spaces in place of some values
aq$Ozone[1] <- ""     # empty string
aq$Ozone[2] <- " "    # space

# Check the first few rows
head(aq, 5)
is.na(aq$Ozone)   # Only original NA values will be TRUE
aq$Ozone == ""           # TRUE for exact empty strings only
trimws(aq$Ozone) == ""   # TRUE for empty strings and spaces
aq$Ozone[trimws(aq$Ozone) == ""] <- NA
is.na(aq$Ozone)
```